

# WASP-52b

R-0506

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-52\_180904 · created 2026-07-20T18:49:42+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458365.8674 +/- 0.0043 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.215 +/- 0.061                 |
| Transit depth (Rp/Rs)^2                           | 4.64 +/- 2.64 [%]               |
| Orbital Inclination (inc)                         | 88.22 +/- 2.8                   |
| Airmass coefficient 1 (a1)                        | 0.85 +/- 0.6                    |
| Airmass coefficient 2 (a2)                        | 0.18 +/- 1.23                   |
| Scatter in the residuals of the lightcurve fit is | 75.76 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 1.11                            |
| Optimal Annulus                                   | 7.5                             |
| Transit Duration (day)                            | 0.086 +/- 0.011                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.215 ± 0.061 vs 0.1646 (deviation 0.83σ) |
| Duration fit vs published | 0.086 d vs 0.0758 d (deviation 0.92σ)     |
| Mid-time O–C              | 10.7 min                                  |
| Depth SNR                 | 3.5                                       |
| Residual scatter          | 75.76 %                                   |

## Light curve

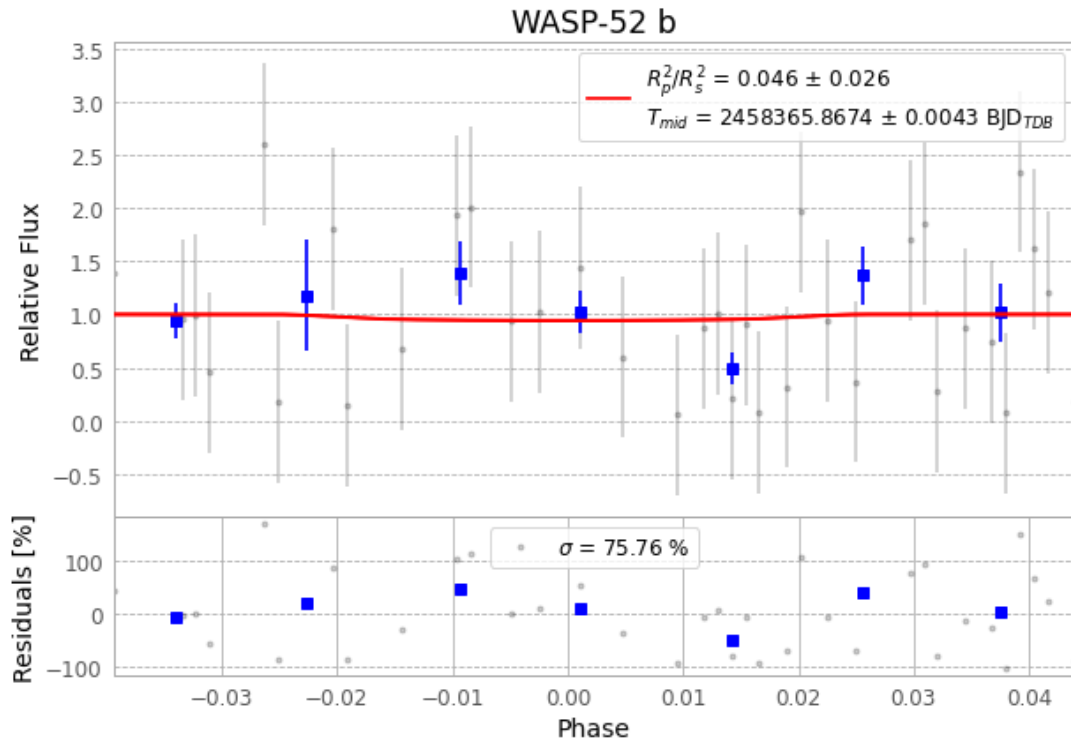


Figure 1 — FinalLightCurve\_WASP-52 b\_2018-09-03.png (SHA-256 d534cf0e0f842f0f...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 210], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 6f93d4cf9208fa9d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 39d9b80

## Data provenance

76 FITS frames (50 MB), WASP-52180904064512.FITS → WASP-52180904103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180904.log (c49a9dcce3db...), FinalLightCurve\_WASP-52 b\_2018-09-03.png (d534cf0e0f84...), FinalLightCurve\_WASP-52 b\_2018-09-03.csv (e540a765ab95...)

## Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)